TCP/IP & Socket Programming Blanks

1. Identify the class of following IP address: 4.5.6.7.

a) Class A b) Class B c) Class C d) Class D

Ans: a) Class A

Explanation: Class A: 1.0.0.0 to 126.255.255.255
Class B: 128.0.0.0 to 191.255.255.255
Class C: 192.0.0.0 to 223.255.255.255
Class D: 224.0.0.0 to 239.255.255.255
Class E: 240.0.0.0 to 255.255.255.255

IP address 4.5.6.7 falls within the range of Class A addresses (1.0.0.0 to 126.255.255.255).

2. The loop-back address is used to send a packet from the ____to ___ on the network.

a) Host; all other hosts b) Router; all other hosts c) Host; specific host d) Host; itself

Ans: d) Host; itself

3. What destination address can be used to send a packet from a host with IP address 188.1.1.1 to all hosts on the network?

a) 188.0.0.0 b) 0.0.0.0 c) 255.255.255.255 d) 188.1.255.255

Ans: c) 255.255.255.255

4. Repeaters function in __ layer of OSI?

a) Physical b) Data link c) Network d) a and b

Ans: a) Physical

5. A bridge can access to the ____address of a station on the same network.

a) Physical b) Network c) Service access point d) All

Ans: a) Physical

Explanation: The physical address, also known as the MAC (Media Access Control) address, uniquely identifies each device on a network segment.

6. As the data packet moves from lower to upper layers, headers are_____.

a) Added b) Subtracted c) Rearranged d) Modified

Ans: a) Added

7. What is the main function of the transport layer?

a) Node-to node delivery b) Process to process message delivery

c) Synchronization d) Updating and maintenance of routing tables

Ans: b) Process to process message delivery

8. IPv6 has ___ bit addresses.

a) 32 b) 64 c) 128 d) Variable

Ans: c) 128

9. When a host on network A sends a message to a host on network B, which address does the router look at?

a) Port b) IP Address c) Physical d) None of the above

Ans: b) IP Address

10. Which of the following is an application layer service?

a) Network virtual terminal b) File transfer, access, and management

c) Mail service d) All of the above

Ans: d) All of the above

11. The checksum in the IP packet covers_____

a) Just the header b) Just the data

c) The header and data d) Just the source and destination address

Ans: c) The header and data

12. Which of the following is not an IPC event?

a) Semaphore b) Message queue c) Shared memory d) Socket

Ans: d) Socket

13. Socket is using

a) Port number b) Interrupt number c) Memory address d) All

Ans: a) Port number

14. Out of these which is a major difference between SOCK_STREAM and SOCK_SEQPACKET

a) one is used for TCP other for UDP b) both are same. No difference at all

c) one with fixed length d) one is sequenced other is non sequenced

Ans: d) one is sequenced other is non sequenced

15. What is advantage of RAW socket

a) easy to program b) fast accessing c) direct interface to IP d) none of these

Ans: c) direct interface to IP

16. What kind of data byte ordering is followed in sockets

a) little endian b) big endian c) depends on hardware d) none of these

Ans: c) depends on hardware

17. which of the following is true about bind()

a) bind is must in client process b) returns non-zero positive integer on success

c) Third argument in bind is a pointer d) as client does not listen to incoming request bind is not necessary

Ans: c) Third argument in bind is a pointer

18. In a TCP connection if sendto is used with valid struct sockaddr pointer in 5th argument then

a) returns error b) no problem data can be passed

c) converts the TCP socket to UDP d) none of these

Ans: a) returns error

19. Which of the following OSI layers is responsible for Identifying communications partners

a) Application b) Session c) Network d) Presentation

Ans: a) Application

20. A broadcast message is an example of which dialog control method

a) Half Duplex b) Base band c) Broad band d) Simplex

Ans: d) Simplex

21. What is a router

a) A hardware device that connects dissimilar networks, such as Cat 5 cabling and FDDI

b) A network host that reads the source and destination addresses in the packet header and makes decisions about where to forward the packet

c) A network host that can forward LAN-based email messages onto the Internet, after repackaging them into the SMTP format

d) A software system that can translate between dissimilar networks such as Ethernet and Token Ring

Ans: b) A network host that reads the source and destination addresses in the packet header and makes decisions about where to forward the packet

22. Which of the following is Class C IP address?

a) 10.10.14.118 b) 135.23.112.57 c) 191.200.199.199 d) 204.67.118.54

Ans: d) 204.67.118.54

23. Which of the following is suitable for mission critical and time sensitive applications?

a) 10BaseT b) 10Base2 c) Token Ring d) Mesh

Ans: c) Token Ring

Explanation: Token Ring networks can guarantee a certain level of performance and ensure that data will be delivered within a predictable timeframe.

24. What is the default subnet mask for Class C network?

a) 255.0.0.0 b) 255.255.0. c) 255.255.255.0 d) 255.0.255.0

Ans: c) 255.255.255.0

25. Which of the following network topologies have each computer connected to a central point?

a) Bus b) Ring c) Star d) Mesh

Ans: c) Star

Explanation: In a star network topology, each computer (or device) is connected directly to a central point, typically a network switch or hub.

26. Which of the following network topologies is the most fault tolerant?

a) Bus b) Mesh c) Star d) Ring

Ans: b) Mesh

27. A Bus network topology is best described as which of the following?

a) All computers are attached to a single cable in a chain style

b) All computers are connected to a central point

c) All computers are connected to every other computer or resource in the network

d) Fault Tolerant

Ans: a) All computers are attached to a single cable in a chain style

28. 10BaseT Ethernet networks, most commonly use which of the following types of cable?

a) Fiber b) coax c) STP d) UTP

Ans: d) UTP

29. A hub works in conjunction with which of the following network topologies?

a) Ring b) Star c) Bus d) Mesh

Ans: b) Star

30. Which of the following is a correct MAC address?

a) 190.168.80.1 b) www.certyourself.com

c) 08:34:FE:4C:F2:54 d) G2:G3:23:87:GG

Ans: c) 08:34:FE:4C:F2:54

Explanation: MAC (Media Access Control) addresses are unique identifiers assigned to network interfaces for communications on the physical network segment. They are usually represented as a series of hexadecimal numbers separated by colons or hyphens.

31. A Hub operates at which of the following layers of the OSI model?

a) Physical b) Session c) Transport d) Application

Ans: a) Physical

32. Which of the following devices operate at the Data Link layer of the OSI model?

a) A Hub b) A NIC c) A Switch d) A Router

Ans: c) A Switch

33.A Router operates in which layer of the OSI model?

a) The Physical Layer b) The Data Link Layer

c) The Transport Layer d) The Network Layer

Ans: d) The Network Layer

34.Which of the following is the correct order of the OSI model from bottom to top?

a) Application, Presentation, Session, Transport, Network, Data Link, Physical

b) Physical, Network, Data Link, Session, Transport, Application, Presentation

c) Physical, Data Link, Network, Transport, Session, Presentation, Application

d) Application, Session, Presentation, Transport, Network, Data Link, Physical

Ans: c) Physical, Data Link, Network, Transport, Session, Presentation, Application

35.Which of the following TCP/IP protocols is used to communicate between a Web Browser and a Web Server?

a) SMTP b) HTTP c) UDP d) POP3

Ans: b) HTTP

36. Which of the following layers of the OSI model does a Bridge operate in?

a) The Data Link Layer b) The Network Layer

c) The Transport Layer c) The Session Layer

Ans: a) The Data Link Layer

37. Which of these protocols provide fast connectionless communications that relies on upper layers of the OSI model for error correction?

a) TCP b) UDP c) SPX d) None

Ans: b) UDP

38. The fastest method of transporting data with a switch?

a) store and forward b) cut through c) loop back d) None

Ans: b) cut through

39. Topology that offers the highest level of redundancy, is easy to troubleshoot but is expensive to install. It is the topology of the internet?

a) star b) ring c) mesh d) bus

Ans: c) mesh

40. OSI layer responsible for mail and file transfers?

a) transport b) data link c) application d) physical

Ans: c) application

41. What is the first octet range for a class A IP address?

a) 1 – 126 b) 1 – 128 c) 1 – 198 d) none

Ans: a) 1 – 126

42. Network connection device that is capable of sending packets along multiple paths depending on which path is the most efficient?

a) bridge b) router c) all the above d) none of the above

Ans: b) router

43. OSI layer that determines whether the transmission is a full or half-duplex transfer?

a) presentation b) session c) application d) physical

Ans: d) physical

44. This OSI layer packages bits into frames?

a) physical b) data – link c) network d) application

Ans: b) data – link

45. OSI layer responsible for error checking frames and adding MAC addresses to its tables?

a) presentation b) transport c) data – link d) network

Ans: c) data – link

46. The initial formation of the packet header is started at this OSI level?

a) Session b) Application c) Physical d) data – link

Ans: d) data – link

47. For a Class C IP address the first ___ octet(s) are reserved for the network ID?

a) 1 b) 2 c) 3 d) 4

Ans: c) 3

48. A set of "rules" for network transmissions?

a) topologies b) protocols c) gateways d) all

Ans: b) protocols

49. What happens when a device is disconnected from the middle of a BUS network?

a) The hub will propagate error messages to the rest of the network

b) All devices on the network will fail

c) The Server will stop responding to that device

d) Data will not be able to get to devices on the other side of this device

e) If a bridge is on the network, only that segment will be effect

Ans: d) Data will not be able to get to devices on the other side of this device

50. IP is responsible for sorting and _____ of packets.

a) Addressing b) Holding c) Organizing d) Delivery

Ans: d) Delivery

51. What is the first octet range for a class B IP address?

a) 127-191 b) 128-191 c) 192-223 d) 1-127

Ans: b) 128-191

52. What is the first octet range for a class C IP address?

a) 1-126 b) 128-191 c) 192-223 d) 192-225

Ans: c) 192-223

53. How many bits are there in an IP address V4?

a) 16 b) 32 c) 64 d) 128

Ans: b) 32

54. A network bridge operates at which layer of the OSI model?

A) Physical b) Data Link c) Network d) Transport

Ans: b) Data Link

55. Each network interface card is uniquely Identified by its?

a) transceiver type b) MAC address c) IP address d) wins address

Ans: b) MAC address

56. The topology most commonly used with a token ring network is the?

a) star b) star – bus c) ring d) star – ring

Ans: c) ring

Explanation: In a token ring network, each workstation or device is connected to the next in a circular manner, forming a closed loop or ring.

57. The primary purpose of a network is?

a) interconnect computers for ease of administration

b) provide central access to workstation hard disks

c) share resources

d) provide increased security for all workstations

Ans: c) share resources

58. The recommended upper limit for the number of workstations that can be included in a peer-peer network is?

a) 5 b) 10 c) 12 d) unlimited

Ans: b) 10

Explanation: recommendation is to keep the number of workstations in a peer-to-peer network relatively small, typically around 10 to 15 devices. Beyond this range, the network may experience degraded performance and reliability issues.

59. An ethernet network is a _____network?

a) active b) passive c) active-passive d) passive-aggressive

Ans: a) active

60. The OSI model layer that formats data packets and provides for error checking and correction is?

a) network b) data link c) transport d) session

Ans: b) data link

61. The network layer device that determines and sends packets are the most efficient route available is a _____?

a) bridge b) switch c) active hub d) router

Ans: d) router

62. packets on the network layer are addressed using?

a) network address b) node address

c) MAC address d) wins address

Ans: a) network address

63. The process that enables faster devices to communicate with slower devices without the loss of data is?

a) error handling b) flow control c) CRC d) parity

Ans: b) flow control

64. On the OSI model layers what is level 6? Ans: Presentation Layer

65. On the OSI model what is level 7? Ans: Application Layer

66. On the OSI model what is level 5? Ans: Session Layer

67. On the OSI model what is level 4? Ans: Transport Layer

68. On the OSI model what is level 3? Ans: Network Layer

69. On the OSI model what is level 2? Ans: Data Link Layer

70. On the OSI model what is level 1? Ans: Physical Layer

71. what hardware devices does the data link layer deal with?

Ans: Network Interface Cards (NICs), switches, bridges

72. what hardware devices does the transport layer deal with?

Ans: Hosts, computers, servers

73. Segmentation of a data stream happen at which layer of OSI model

a) physical b) datalink c) network d) transport

Ans: d) transport

74. When data is encapsulated, which is the correct order

a) data, frame, packet, segment, bit b) segment, data, packet, frame, bit

c) data, segment, packet, frame, bit d) data, segment, frame, packet, bit

Ans: c) data, segment, packet, frame, bit

75. Which of the following would describe a transport layer connection that would ensure reliable delivery

a) Routing b) acknowledgement c) switching d) system authentication

Ans: b) acknowledgement

76. What is a reason you want to use switches in your network instead of hub?

a) They're inexpensive b) switches are faster than hubs at reading frames

c) switches create more collision domains d) switches do not forward broadcast.

Ans: b) switches are faster than hubs at reading frames

77. What protocol is used to find the hardware address of a local device

a) RARP b) ARP c) ICMP d) IP

Ans: b) ARP

78. What is the result of using a hierarchical addressing scheme?

a) Increased number of addresses b) decreased number of routers needed

c) increased memory usage on routers d) no routing tables needed on routers

Ans: b) decreased number of routers needed

79. If you wanted to have 12 subnet using class c net ID which subnet mask would you use?

a) 255.255.255.252 b) 255.255.255.242 c) 255.255.255.240 d) 255.255.255.255

Ans: c) 255.255.255.240

80. Which of the following statement is true?

a) TCP is connection oriented but doesn't use flow control

b) IP is not necessary on all hosts that use TCP

c) ICMP must be implemented by all TCP/IP hosts is False

d) ARP is used to find IP address from known hardware address is False

Ans: c) ICMP must be implemented by all TCP/IP hosts is False

81. Which class of IP address provides maximum of only 254 host address per network ID

a) class A b) Class B c) Class C d) Class D

Ans: c) Class C

82. Which protocol uses ICMP?

a) telnet b) ping c) ARP d) IP

Ans: b) ping

83. Maximum port number allowed in socket is

a) up to 4 GB b) up to 65535 c) no limit d) 255

Ans: b) up to 65535

84. To use the UDP socket for communication

a) IP address is not essential b) Port number is optional

c) a and b d) None

Ans: a) IP address is not essential

85. Accept system call returns

a) a new socket ID on success b) replaces the existing socket ID on failure

c) a and b d) none

Ans: a) a new socket ID on success

86. What is the default subnet mask of the IP 10.0.0.10?

Ans: Class A

Explanation: The IP address 10.0.0.10 falls within the range of Class A addresses (1.0.0.0 to 126.0.0.0), the default subnet mask would be 255.0.0.0.

87. Which of the following service defines the responsibility of presentation layer

a) format & code conversion b) establish, manage and terminate session

c) ensures reliability of transmission of data d) log-in & log-out procedures

Ans: a) format & code conversion

88. In LAN, using IEE 802.3 Ethernet protocol, what is the topology followed?

a) can be either bus or star b) must be logical bus

c) must be physical bus d) must be logical star

Ans: a) can be either bus or star

89. The destination address field in 802.3 frame is of _____bytes.

a) 2 b) 4 c) 6 d) 8

Ans: c) 6

90. A bridge used inside a LAN can read incoming

a) frame b) datagram c) both d) none

Ans: a) frame

91. A layer-2 switch works by using

a) MAC address b) IP address c) Subnet address d) none

Ans: a) MAC address

92. For a host to communicate in LAN / WAN, the IP address

a) must be unique b) can be dummy c) is optional d) not required

Ans: a) must be unique

93. Which of the following is true about ARP?

a) request is broadcast but reply is unicast b) request is unicast but reply is broadcast

c) request and reply both are unicast d) request and reply both are broadcast

Ans: a) request is broadcast but reply is unicast

94. The RARP packet in a host _____

a) knows its physical address b) knows its IP address c) a & b d) none

Ans: b) knows its IP address

95. The maximum total size and maximum fragment size of IP datagram is

a) 65,535 bytes and depends on MTU by frame b) 65,535 and 65,535

c) 65535 & 576 d) none of the above

Ans: c) 65535 & 576

96. What is the maximum size of data can be stuffed into UDP?

a) 1024 bytes b) 64KB c) 512 bytes d) Any size

Ans: d) Any size

97. What are the port numbers of the following SMTP, TFTP, FTP?

a) 21,25,69 b) 25,69,21 c) 69,21,25 d) none

Ans: b) 25,69,21

98. What is the responsibility of TCP?

a) Multiplexing b) reliability c) Flow control d) all

Ans: b) reliability

99. Which function is used to communicate data in connectionless Host

a) send to b) send c) write d) any of the above

Ans: d) any of the above

100. htons() function returns

a) 16bit integer in network byte order b) 16bit integer in host byte order

c) a or b d) none

Ans: a) 16bit integer in network byte order

101. send() function returns

a) -1 on error b) no. of bytes sent on success c) a & b d) none

Ans: c) a & b

102. What is the second argument in 'connect()'.

a) struct sockkaddr * b) struct sockaddr c) sizeof(struct sockaddr) d) none

Ans: b) struct sockaddr

103. When the incoming request queue created by a listen in a server is "listen (socket ID, 11)", then which of the following statement is true?

a) If 10 active clients are communicating then the 11th client's connection request fails.

b) If 11 clients are actively connected then 12th client's connection request fails.

c) 100 users concurrently can use the server.

d) None of these.

Ans: b) If 11 clients are actively connected then 12th client's connection request fails.

104. What is the default subnet mask for Class A network?

a) 255.0.0.0 b) 255.255.0.0 c) 255.255.255.0 d) 255.0.255.0

Ans: a) 255.0.0.0

105. UDP offers which of the following benefits relative to TCP?

a) UDP consumes fewer computer resources by not maintaining connection state

b) UDP supports a self-regulating "throttle" feature that prevents network saturation

c) UDP guarantees that individual packets of a transmission will arrive "in order"

d) none of the above

Ans: a) UDP consumes fewer computer resources by not maintaining connection state

106. TCP offers which of the following benefits relative to UDP?

a) TCP allows much larger packets to be sent over the network, improving application performance

b) TCP includes a "retry" mechanism to prevent missing (dropped) packets

c) TCP uses a smaller header than does UDP

d) none of the above

Ans: b) TCP includes a "retry" mechanism to prevent missing (dropped) packets

107. The size of a UDP header is _____

a) 2 bytes b) 4 bytes c) 8 bytes d) 16 bytes

Ans: c) 8 bytes

108. In the current version of IP (IPv4), the use of TCP and UDP headers differ in which of the following ways?

a) TCP checksums are optional but UDP checksums are mandatory

b) UDP checksums are optional but TCP checksums are mandatory

c) TCP checksums are larger than UDP checksums

d) UDP checksums are larger than TCP checksums

Ans: b) UDP checksums are optional but TCP checksums are mandatory

109. The so-called "well-known" TCP and UDP port numbers fall into which numeric range?

a) 0 to 999 b) 0 to 1000 c) 0 to 1023 d) 0 to 1024

Ans: c) 0 to 1023

110. The so-called "dynamic" TCP and UDP port numbers fall into which numeric range?

a) 8192 to 49151 b) 31768 to 54220 c) 49152 to 65535 d) 54220 to 65535

Ans: c) 49152 to 65535

111. What is the size of next version IP(IPv6)?

a) 4 bytes b) 8 bytes c) 16 bytes d) 32 bytes

Ans: c) 16 bytes

112. Which of the following IPv4 addresses is the loopback address?

a) 0.0.0.0 b) 10.0.0.1 c) 127.0.0.1 d) 255.255.255.255

Ans: c) 127.0.0.1

113. The length of an IP datagram header can vary depending on the options used) What is the smallest possible IP header size?

a) 16 bytes b) 20 bytes c) 24 bytes d) 60 bytes

Ans: b) 20 bytes

114. The source IP address for an IPv4 datagram is stored where in the header?

a) 2nd longword (bytes 5-8) b) 3rd longword (bytes 9-12)

c) 4th longword (bytes 13-16) d) 5th longword (bytes 17-20)

Ans: a) 2nd longword (bytes 5-8)

115. The datagram length field in an IPv4 datagram contains which numeric value?

a) total number of bytes in the datagram, including the header

b) total number of bytes in the datagram, excluding the header

c) total number of longwords in the datagram, including the header

d) total number of longwords in the datagram, excluding the header

Ans: b) total number of bytes in the datagram, excluding the header

116. The maximum size of an IP datagram (including the header) is _____

a) 1,024 bytes b) 1,500 bytes c) 4,096 bytes d) 65,535 bytes

Ans: d) 65,535 bytes

117. What function does Address Resolution Protocol (ARP) perform?

a) converts IP addresses to Ethernet addresses

b) automatically assigns IP addresses to computers

c) prevents two computers from using the same IP address

d) none of the above

Ans: a) converts IP addresses to Ethernet addresses

118. Which utility program is designed to report whether a networked computer is responding at a given IP address, and how long these responses take?

a) traceroute b) ping c) TTCP d) netstat

Ans: b) ping

119. Which of these shell commands supplies the IP address of a computer running the Windows operating system?

a) ipshow b) traceroute c) iplookup d) ipconfig (or winipcfg)

Ans: d) ipconfig (or winipcfg)

120. Each of the four numeric fields of an IP address written in dotted-decimal notation is also known as _____

a) octet b) word c) subnet d) class

Ans: a) octet

121. The Class D network range is reserved on the Internet for _____

a) broadcast b) no cast c) anycast d) none of the above

Ans: d) none of the above

Explanation: Class D network range is reserved for multicast addresses. Multicast allows a device to send a single packet to multiple destinations.

122. A Class C network can handle a maximum of how many hosts?

a) 254 b) 258 c) 510 d) 512

Ans: a) 254

123. Which central authority is responsible for allocating IP addresses on the Internet?

a) IETF b) IANA c) IEEE d) none of the above

Ans: b) IANA

124. Which Request For Comments (RFC) documentation describes the Internet Protocol?

a) RFC 1 b) RFC 768 and RFC 1157

c) RFC 761 d) RFC 760 and RFC 791

Ans: d) RFC 760 and RFC 791

125. The IP checksum feature supports integrity checking for _____

a) an IP packet (header and payload) b) an IP header only

c) an IP payload only d) none of the above - IP doesn't use checksums

Ans: a) an IP packet (header and payload)

126. The IP delivery service is _____

a) reliable, connection-oriented b) non-routable

c) unreliable, connectionless d) none of the above

Ans: c) unreliable, connectionless

127. MTU stands for _____

a) Minimum Transfer Unit b) Minimum Transmission Unit

c) Maximum Transmission Unit d) Maximum Transfer Unit

Ans: c) Maximum Transmission Unit

128. Which of these network devices belong at the OSI data link layer (layer two)?

a) router b) bridge c) VPN d) none of the above

Ans: b) bridge

129. Which of these network devices primarily functions at the OSI Network layer (layer 3)?

a) switch b) gateway c) router d) all of the above

Ans: c) router

130. Which Protocol Data Unit (PDU) is employed at the Transport Layer?

a) bits b) frames c) packets d) segments

Ans: d) segments

131. What layer of the OSI model is associated with the MAC and LLC sublayers?

a) physical b) data link c) network d) transport

Ans: b) data link

132. what is out-of-band data

a) urgent data b) more data than band-width c) data in disorder d) none of these

Ans: a) urgent data

133. If bind( ) fails on a socket what is the return

a) nothing b) 0 c) -1 d) None

Ans: c) -1

134. After bind( ) failure , what is the best operation to complete process

a) call exit b) use return c) use close( ), then close the process d) any of these

Ans: c) use close( ), then close the process

135. stream sockets are

a) Half duplex b) Simplex c) Full duplex d) None

Ans: c) Full duplex

136. why bind( ) function not used in client side

a) because connect function serves the purpose b) can be used

c) to avoid extra connection d) none

Ans: c) to avoid extra connection

137. what is the loop back address for the system with IP 192.168.1.21

a) 192.168.1.255 b) 0.0.0.0 c) 127.0.0.1 d) None

Ans: c) 127.0.0.1

138. what is the broad-cast address for 1st subnet, with IP 192.168.1.21 and mask as255.255.255.192

a) 255.255.255.255 b) 255.255.255.0 c) 255.255.255.192 d) None

Ans: c) 255.255.255.192

139. sendto( ) is used

a) in connection mode b) in connection-less mode c) raw socket mode d) none

Ans: b) in connection-less mode

140. Packets in the IP layer is called _____. Ans: IP Packets or Datagrams

141. Given the IP address 126.110.16.7 what class of address is it? Ans: Class A

142. The error detection method used by IP is _____. Ans: Checksum

143. Router works in how many layers? What are they?

Ans: Routers work in three layers of the OSI model. They are

1. Network Layer (Layer 3)
2. Data Link Layer (Layer 2)
3. Physical Layer (Layer 1).

144. Which of the following statements is true?

a) A switch supports more ports than a bridge.

b) In switches each individual port could be configured for different data rate.

c) Both

d) None of the this

Ans: c) Both

145. What is the Broadcast address of a system with IP as 191.0.1.191?

Ans: 191.0.1.255

Explanation: The broadcast address of a system with IP address as 191.0.1.191 with a subnet mask of 255.255.255.0 is 191.0.1.255.

146. What is the subnet mask for a IP address 192.168.1.2 with 10 subnets?

Ans: 255.255.240.0.

Explanation: 192.168.1.2 -> 11000000.10101000.00000001.00000010
To create 10 subnets, borrow 4 bits from the host portion, making the subnet mask:
11111111.11111111.11110000.00000000 (Converting this back to decimal notation)
255.255.240.0
The subnet mask for creating 10 subnets from the IP address 192.168.1.2 is 255.255.240.0.

147. Ethernet is also known as?

a) 802.5 b) 803.2 c) 802.11 d) 802.3

Ans: b) 803.2

148. In a Basic Ring topology, if there is a line break, or if user is adding or removing a device anywhere in the ring this will bring down the network. (True / False).

Ans: True

149. Mesh topologies are the least expensive and are easier to maintain. (True / False).

Ans: False

150. If computers are connected along a single cable this is called what type of topology?

Ans: Bus Topology

151. The no. of computers on a bus network will NOT affect network performance. (True / False).

Ans: False

Explanation: The number of computers on a bus network affects network performance because all data transmitted on the bus is received by every device, leading to potential congestion and collisions.

152. What is the size of an IP fragment?

Ans: The size of an IP fragment depends on the Maximum Transmission Unit (MTU) of the network over which it is transmitted. IP fragments are created when an IP datagram is too large to fit in the MTU of a network segment and must be divided into smaller fragments for transmission.

153. How many bytes is the sequence number field in TCP Header?

Ans: 4 bytes or 32 bits

154. ARP implemented in a switch. (TRUE / FALSE).

Ans: FALSE

Explanation: ARP (Address Resolution Protocol) is not implemented in a switch. It is typically implemented in hosts and routers to map IP addresses to MAC addresses.

155. IP fragment carries IP Header. (TRUE/ FALSE).

Ans: FALSE

Explanation: IP fragments carry the IP payload (data) along with the IP header.

156. List out the flags used in TCP Header.

Ans: The flags used in the TCP header are:

1. URG (Urgent)
2. ACK (Acknowledgment)
3. PSH (Push)
4. RST (Reset)
5. SYN (Synchronize)
6. FIN (Finish)

157. What is the return value of gethostbyname() on success ?

Ans: Pointer to a hostent structure.

158. For a Class B IP address the first ___ octet(s) are reserved for the network ID?

Ans: 2 octets

159. What is the address of limited Broadcast address?

Ans: 255.255.255.255

160. Which Protocol Data Unit (PDU) is employed at the Data Link Layer?

Ans: Frame

161. Which Protocol Data Unit (PDU) is employed at the Network Layer?

Ans: Packet

162. Which Protocol Data Unit (PDU) is employed at the physical Layer?

Ans: Bit

163. What are the smallest and largest possible values for an IP octet.

Ans: 0 and 255

164. How to convert a string format of IP address into an protocol specific internet address?

Ans: inet_pton() function from the arpa/inet.h header file.

165. Which of the following statements is(are) true

a) TFTP uses UDP services
b) TFTP has a flow-control mechanism
c) TFTP has five types of messages
d) all the above.

Ans: a) TFTP uses UDP services

Explanation: TFTP (Trivial File Transfer Protocol) uses UDP services, does not have a flow-control mechanism, and has five types of messages.

166. A bridge filters traffic using which type of address?

Ans: MAC (Media Access Control) address.

167. What topology provides each device with a point –to – point connection to every other device in the network?

Ans: Mesh Topology

168. What must be configured in TCP/IP for two computers on the same segment to communicate with each other?

a) IP address and subnet mask b) DNS address c) all of the above d) none

Ans: a) IP address and subnet mask

169. For a Class A IP address the first ___ octet(s) are reserved for the network ID?

a) 1 b) 2 c) 3 d) 4

Ans: a) 1

170. What is the subnet broadcast address for Ip 192.168.1.100 with subnet mask as 255.255.255.224?

Ans: 192.168.1.127.

Explanation: IP address: 11000000.10101000.00000001.01100100 (192.168.1.100)
Subnet mask: 11111111.11111111.11111111.11100000 (255.255.255.224)
Network address: 11000000.10101000.00000001.01100000
To calculate the broadcast address, flip all host bits in the network portion of the network address to 1.
In this case, the host bits are the last 5 bits of the network address: 00000.
Flipping these bits to 1, we get 11111 (Flipping the host bits means changing any 0 bits to 1.)
The network address in binary with the flipped host bits: 11000000.10101000.00000001.01111111.
(192.168.1.127.)

171. When data is transmitted from device A to device B, the header from A's layer 5 is read by B's _____ layer.

a) physical b) transport c) session d) presentation

Ans: b) transport

172. Session layer checkpoints _____.

a) allow just a portion of a file to be resent b) detect and recover errors

c) control the addition of headers d) are involved in dialog control

Ans: d) are involved in dialog control

173. Encryption and decryption are functions of the _____ layer.

a) transport b) session c) presentation d) application

Ans: c) presentation

174. Match the following to one of the seven OSI layers:

a) reliable end–to–end data transportation — Ans: Transport Layer

b) network selection — Ans: Network Layer

c) defines frames — Ans: Data Link Layer

d) provides user services such as e-mail and file transfer — Ans: Application Layer

e) transmission of bit stream across physical medium — Ans: Physical Layer

175. A Packet from an Ethernet network requires a _____ before it can be routed to an FDDI network.

a) repeater b) bridge c) router d) gateway

Ans: c) router

176. Identify the following IP address: 169.5.0.0.

a) host IP address b) direct broadcast address

c) limited broadcast address d) network address.

Ans: d) network address

177. Using the direct broadcast address, a_____ sends a packet to _____ on the network.

a) Host; all other hosts b) Router; all other hosts

c) Host; specific host d) Host; itself

Ans: a) Host; all other hosts

178. Using the limited broadcast address, a_____ sends a packet to _____ on the network.

a) Host; all other hosts b) Router; all other hosts

c) Host; specific host d) Host; itself

Ans: a) Host; all other hosts

179. Find the class of the following IP addresses.

a) 208.34.54.12 Ans: Class C

b) 238.34.2.1 Ans: Class D

c) 114.34.2.8 Ans: Class A

d) 129.14.6.8 Ans: Class B

e) 241.34.2.8 Ans: Class E

180. Find the netID and hosted of the following IP addresses:

a) 192.8.56.2 Ans: NetID – 192.8.56, HostID - 2

b) 220.34.8.9 Ans: NetID – 220.34.8, HostID – 9

c) 208.34.54.12 Ans: NetID – 208.34.54, HostID – 12

d) 205.23.67.8 Ans: NetID – 205.23.67, HostID – 8

181. A device has the IP address 190.1.2.3. What is its subnet ID?

a) 1 b) 2 c) 3 d) insufficient information

Ans: d) insufficient information

182. Which of the following is the default mask for the address 198.0.46.201?

a) 255.0.0.0 b) 255.255.0.0 c) 255.255.255.0 d) 255.255.255.255

Ans: c) 255.255.255.0

183. What class of IP address does the subnet mask 255.128.0.0 operate on?

a) A b) B c) C d) A or B or C

Ans: a) A

Explanation:

- Class A addresses have their first octet in the range 1-126, with a default subnet mask of 255.0.0.0.
- Class B addresses have their first octet in the range 128-191, with a default subnet mask of 255.255.0.0.
- Class C addresses have their first octet in the range 192-223, with a default subnet mask of 255.255.255.0.

184. A subnet mask of 255.255.255.248 allows how many host IDs per subnet? (Disregard special addresses).

a) 3 b) 6 c) 8 d) 248

Ans: b) 6

Explanation: Number of Hosts=2^(Number of Host Bits) −2 = 2^(3)-2= 8-2= 6

185. A subnet mask of 255.255.255.224 allows how many host IDs per subnet? (Disregard special addresses).

a) 3 b) 7 c) 8 d) 30

Ans: d) 30

Explanation: Number of Host Bits=32-27=5

Number of Hosts=2 ^(Number of Host Bits) −2 = 2^(5)-2 = 32-2 = 30

186. The subnet mask for a class C network is 255.255.255.192. How many subnet works are available? (Disregard special addresses).

a) 2 b) 4 c) 8 d) 192

Ans: b) 4

Explanation: 2^2 = 4 subnets

187. The subnet mask for a class B network is 255.255.255.192. How many subnet works are available? (Disregard special addresses).

a) 2 b) 4 c) 1,024 d) 192

Ans: c) 1,024

Explanation: the default subnet mask for Class B is 255.255.0.0. (/16) to make 255.255.255.192. (/26)
Number of bits borrowed for subnetting=10 (26-16)
Number of subnets=2^ (number of bits borrowed for subnetting)
Number of subnets = 2^ (10) = 1024
Number of subnets = 1024

188. The IP address of a host on a class C network is 198.123.46.237. Four subnets are allowed for this network. What is the subnet mask? Assume a contiguous subnet address space.

Ans: 255.255.255.192.

Explanation: to accommodate four subnets, need to borrow at least two bits (2^2 = 4 subnets). By borrowing two bits for subnetting, changing the default subnet mask from 255.255.255.0 (/24) to 255.255.255.192 (/26) in binary (11111111.11111111.11111111.11000000.).

189. What is subnetwork address and subnet mask for the IP address 142.45.43.44? Assume a contiguous subnet address space.

Ans: 142.45.43.0. and 255.255.255.192

Explanation: Assume to create 4 subnets, need to borrow 2 bits from the host portion to create the subnet portion (2^2 = 4). With 2 bits borrowed for subnetting, the new subnet mask will be 255.255.255.192.

IP address: 10001110.00101101.00101011.00101100 (142.45.43.44)
Subnet mask: 11111111.11111111.11111111.11000000 (255.255.255.192)
Subnetwork address: 10001110.00101101.00101011.00000000 (142.45.43.0.)

190. What is the range of hosts for the subnetwork with an address of 150.20193.4? Assume a contiguous subnet address space.

Ans: range of host is 0-255

191. Find the class of the following subnetworks:

a) 22.14.0.0 Ans: Class A

b) 133.45.6.0 Ans: Class B

c) 16.14.9.0 Ans: Class B

d) 193.45.127.64 Ans: Class C

192. Find the subnetwork address for the following:

a) IP Address: 125.34.12.56 Ans: 125.34.0.0

Mask: 255.255.0.0

b) IP Address: 120.14.22.16 Ans: 120.14.0.0

Mask: 255.255.128.0

c) IP Address: 140.11.36.22 Ans: 140.11.36.0

Mask: 255.255.255.0

d) IP Address: 141.181.14.16 Ans: 141.181.0.0

Mask: 255.255.224.0

e) IP Address: 200.34.22.156 Ans: 200.34.22.144

Mask: 255.255.255.240

Explanation: For the IP address 125.34.12.56 and the subnet mask 255.255.0.0:

IP address in binary: 01111101.00100010.00001100.00111000

Subnet mask in binary: 11111111.11111111.00000000.00000000

Bitwise AND operation: 01111101.00100010.00000000.00000000 (125.34.0.0)

193. In _____ delivery, both the deliverer of the IP packet and the destination are on the same network.

a) a connection less b) a connection-oriented c) a direct d) an indirect

Ans: c) a direct

194. In _____ delivery, both the deliverer of the IP packet and the destination are on the different network.

a) a connection less b) a connection-oriented c) a direct d) an indirect

Ans: d) an indirect

195. In _____delivery, packets of a message are logically connected to one another.

Ans: Connection-oriented

196. A best effort delivery service such as IP does not include _____.

a) error checking b) error correction c) datagram acknowledgement d) all the above

Ans: b) error correction

197. What is the value of the total length field if the header is 28 bytes and the data field is 400 bytes?

a) 428 b) 407 c) 107 d) 427

Ans: a) 428

Explanation: Total length = Header size + Data field size
Total length = 28 bytes + 400 bytes
Total length = 428 bytes

198. The _____module takes fragments of a message and puts them back in order.

a) processing b) routing c) fragmentation d) reassembly

Ans: d) reassembly

199. Which fields of the IP header change from router to router?

Ans: TTL (Time To Live), Checksum

200. What is the first octet range for a class E IP address?

Ans: 240 - 255

2. Which protocol does the fragmentation?

Ans: IP (Internet Protocol)